PRAXIS ENGINEERING PLATFORM • INDUSTRY READINESS SERIES

Data Analyst & Analytics Engineering Handbook

Modern Data Stack, SQL Optimization, ETL Pipelines, Data Contracts & Executive BI Dashboards

A definitive field reference for junior and transitioning Data Analysts and Analytics Engineers navigating modern data warehouses, high-performance SQL, dbt data modeling, automated ETL/ELT pipelines, data governance, and executive business intelligence.

Curriculum Track:	Data Analyst & Analytics Engineering Handbook
Audience:	Junior Engineers, Transitioning Developers, Technical Interns
Platform Version:	Praxis Workplace Simulation v2.4 (Enterprise Edition)
Core Assessment:	5 MEQ Knowledge Check & 40 Production Engineering Problems
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NOTICE TO LEARNERS: This handbook serves as the foundational text for the Praxis Workplace Simulation. Thorough review of all 10 chapters is required prior to taking the 5-Question Readiness Assessment and accessing the 40 Production Problems (20 Basic with Help Statements + 20 Intermediate).

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Learning Invariant: *The 5-question knowledge check draws questions directly from these 10 chapters. Each syllabus topic corresponds to practical scenarios in the 40 production problems.*

CHAPTER 1: DATA TEAM STRUCTURE, BUSINESS STAKEHOLDERS & DECISION SUPPORT

How data practitioners partner with product, marketing, finance, and engineering.

In high-performing data organizations, Data Analysts do not function as 'ad-hoc query vending machines'. Analysts act as strategic thought partners who translate fuzzy business queries into rigorous statistical hypotheses and actionable executive metrics. Analysts collaborate with Data Engineers (who build raw ingest pipelines) and Analytics Engineers (who build dbt transformation models).

FIELD STANDARD: THE DATA REQUEST RULE

When a stakeholder asks for 'a dump of all user transactions in Excel', probe for the underlying decision: 'What business problem are you trying to solve? Let's design a dashboard answering that directly.'

Role	Primary Domain	Critical Interface with Analysts
VP of Product / Business	Revenue growth, retention, OKRs & strategic priorities	Executive summary dashboards, cohort retention analysis
Data Engineer	Ingestion pipelines, raw Kafka/Spark streams, warehouse uptime	Raw data schemas, API ingestion SLAs, data lake partitioning
Analytics Engineer	dbt transformation models, clean dimensional data marts	Testing contracts, star schemas, reusable metric definitions
Data Analyst	Exploratory analysis, SQL queries, dashboard design, A/B testing	Translating data insights into operational business recommendations
Machine Learning Engineer	Predictive algorithms, recommendation models, feature stores	Feature validation, training dataset labeling, target metric definitions

CHAPTER 2: ANALYTICS AGILE, SPRINT PLANNING & MANAGING AD-HOC REQUESTS

Balancing long-term data warehouse modeling against urgent stakeholder requests.

A major challenge in data analytics is managing the onslaught of ad-hoc 'urgent' queries from marketing, finance, and executives. Elite data teams protect their capacity by implementing a 70/30 resource allocation: 70% of sprint capacity is committed to planned roadmap projects (e.g. churn model, self-serve data marts), while 30% is allocated to an 'On-Call Data Triage' rotation for immediate business requests.

FIELD STANDARD: AD-HOC TICKET PROTOCOL

Every incoming ad-hoc data request must have an assigned business impact rating and target decision deadline. Unprioritized Slack messages are routed through the team Jira backlog.

Ceremony	Frequency & Focus	Data Analyst's Deliverable
Data Sprint Planning	Every 2 weeks (60 mins)	Estimate data modeling and dashboard tickets; assign ad-hoc rotation
Daily Data Standup	Daily (15 mins)	Review active analyses, identify upstream data quality blocks
Metric Definition Review	Monthly (45 mins)	Cross-functional consensus on core metric formulas (e.g. definition of 'Active User')
Executive Insights Briefing	Monthly (30 mins)	Present key trends, anomaly findings, and strategic recommendations to leadership
Data Retrospective	End of Sprint (45 mins)	Analyze pipeline breakages, dashboard adoption rates, and ad-hoc volume

CHAPTER 3: VERSION-CONTROLLED SQL, DBT (DATA BUILD TOOL) & GIT WORKFLOWS

Applying software engineering rigor to data transformations.

Analytics code belongs in Git, not floating in unversioned BI query tabs. dbt (data build tool) enables analysts to write modular, reusable SQL select statements with built-in version control, lineage graphing, and automated testing. Every query runs through pull request reviews where data schemas, column naming conventions, and query performance are validated before deploying to production.

FIELD STANDARD: THE SINGLE SOURCE OF TRUTH INVARIANT

Never compute core business metrics (e.g. Gross Merchandise Value) with custom inline SQL in multiple dashboards. Define the metric once in a centralized dbt model and reference it everywhere.

dbt Concept	Function & Mechanism	Data Analyst Best Practice
Staging Models (stg_)	Clean, rename, and cast raw source table fields	1-to-1 with raw source tables; zero business joins or aggregations here
Intermediate Models (int_)	Complex joins, business logic calculations, window functions	Decompose intricate transformations into testable modular steps
Marts Models (fct_ / dim_)	Dimensional star schemas optimized for BI consumption	Fast querying; documented business definitions for every column
Generic Schema Tests	Automated validations: not_null, unique, accepted_values	Run on every CI pull request to prevent duplicate keys or missing foreign IDs

CHAPTER 4: THE MODERN DATA STACK: DATA WAREHOUSES & DIMENSIONAL MODELING

Architecting Snowflake, BigQuery, and Databricks with Kimball Star Schemas.

Modern Cloud Data Warehouses (Snowflake, BigQuery, Amazon Redshift) employ columnar storage architectures designed for high-throughput analytical scans. Analysts structure data using Kimball Dimensional Modeling: Fact Tables (representing numeric business events: orders, clicks, payments) and Dimension Tables (representing context: users, products, dates).

FIELD STANDARD: COLUMNAR STORAGE PERFORMANCE RULE

*Never run 'SELECT *' on production tables with hundreds of columns and billions of rows. Only select the exact columns needed to minimize data scan costs and execution times.*

Data Modeling Component	Characteristics & Purpose	Example Columns
Fact Table (fct_orders)	Narrow, deep table containing measurable numeric business events and foreign keys	order_id, user_id, product_id, order_timestamp, amount_cents, discount_cents
Dimension Table (dim_users)	Wide, descriptive table containing context and attributes	user_id, signup_date, acquisition_channel, country_code, subscription_tier
Slowly Changing Dimension (SCD-2)	Tracks historical attribute changes over time using validity dates	user_id, plan_name, is_current, valid_from_timestamp, valid_to_timestamp
Aggregate Rollup Mart	Pre-calculated daily or monthly summaries for executive reporting	report_date, department_id, total_revenue, active_headcount, gross_margin_pct

CHAPTER 5: HIGH-PERFORMANCE SQL MASTERY: WINDOW FUNCTIONS & QUERY TUNING

Writing clean, efficient, and cost-effective analytical SQL queries.

Advanced SQL is the core technical weapon of the data analyst. Mastery of Window Functions (ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD, SUM OVER PARTITION) enables cohort analysis, running totals, and deduplication without costly self-joins. Analysts must understand execution query plans (EXPLAIN), partition pruning, and cluster keys to avoid table scan timeouts.

FIELD STANDARD: THE SAFE DEDUPLICATION PATTERN

Use ROW_NUMBER() OVER (PARTITION BY user_id ORDER BY updated_at DESC) WHERE row_num = 1 to guarantee deterministic, exact deduplication across mutable source logs.

SQL Technique	Syntax Pattern	Common Analytics Use Case
LAG / LEAD	LAG(revenue, 1) OVER (PARTITION BY store_id ORDER BY month)	Computing Month-over-Month (MoM) revenue growth percentages
Running Total	SUM(amount) OVER (PARTITION BY user_id ORDER BY created_at)	Calculating cumulative customer lifetime value (LTV) progression
Cohort Grouping	MIN(order_date) OVER (PARTITION BY user_id)	Establishing first purchase date for cohort retention heatmaps
Partition Pruning	WHERE date_partition >= CURRENT_DATE - INTERVAL '30 days'	Scanning only relevant storage micro-partitions; reducing query cost by 90%

CHAPTER 6: DATA QUALITY, TESTING CONTRACTS & OBSERVABILITY

Detecting anomalies, silent pipeline corruptions, and schema drift.

The worst data failure is not a crashed pipeline; it is a silently corrupted pipeline that completes successfully and populates executive dashboards with false numbers. Data Observability monitors four core pillars: Freshness (is the data up to date?), Volume (did we receive the expected row count?), Schema (did a column name or type change?), and Distribution (are null rates or averages anomalous?).

FIELD STANDARD: UPSTREAM SCHEMA CONTRACT

Software engineers must never alter production database column names or types without notifying data teams. Enforce Schema Contracts in CI to prevent broken downstream pipelines.

Data Failure Mode	Root Cause	Automated Detection Defense
Stale Executive Dashboard	Airflow ETL job hung without alerting on-call	Data Freshness SLA alerts (warn if max(created_at) > 4 hours old)
Duplicate Revenue Numbers	Upstream retry triggered duplicate webhook ingest	Unique primary key constraint tests running on dbt models
Sudden 0% Conversion Rate	Frontend event tracking SDK was dropped in mobile release	Volume anomaly detection alerting on drop in tracked events/hour
Corrupted Currency Calculations	Upstream API changed response from USD to Cents	Range check tests: assert transaction_amount between $0.01 and $10,000

CHAPTER 7: A/B TESTING, EXPERIMENTATION & STATISTICAL RIGOR

Distinguishing true business signals from statistical noise.

Modern product teams rely on controlled A/B experimentation to validate feature changes. Analysts must enforce rigorous statistical methodology: calculating required sample sizes before launch, establishing statistical power (typically 80%) and significance (alpha = 0.05), and guarding against common pitfalls like p-hacking or early stopping.

FIELD STANDARD: THE PEEKING PROBLEM INVARIANT

Never declare an A/B test winner after 2 days just because p < 0.05. Tests must run for full business cycle iterations (minimum 14 days) to capture weekday vs weekend user variances.

Experimentation Term	Mathematical Definition	Practical Business Meaning
Null Hypothesis (H0)	The assumption that the new feature has no effect on the target metric	Baseline assumption that treatment and control have identical performance
P-Value	The probability of observing results as extreme assuming H0 is true	Threshold for statistical significance (p < 0.05 indicates < 5% chance of false positive)
Sample Size Determination	Calculated based on Minimum Detectable Effect (MDE) and variance	Ensures enough users enter the test to detect meaningful revenue changes
Sample Ratio Mismatch (SRM)	Chi-square goodness-of-fit test comparing observed vs expected traffic split	Detects engineering bugs where users are dropped or assigned unequally (e.g. 48/52 split)

CHAPTER 8: DATA GOVERNANCE, SECURITY, PII MASKING & COMPLIANCE

Protecting customer privacy under GDPR, CCPA, and enterprise security standards.

Data practitioners hold access to the organization's most sensitive customer records. Analysts must strictly adhere to data governance standards: Personally Identifiable Information (PII) such as passwords, credit cards, social security numbers, and email addresses must be masked or hashed before reaching analytical layers. Access to raw production tables is governed by Role-Based Access Control (RBAC).

FIELD STANDARD: THE PII INVARIANT

Never export customer names, phone numbers, or credit card digits to local CSV files or unencrypted spreadsheets. Always use anonymized customer surrogate keys (e.g. user_hash).

Data Classification	Examples	Mandatory Storage & Query Policy
Restricted / PII	Customer name, physical address, national ID, phone number	Column-level encryption; masked by default in all analytics queries
Confidential	Transaction amounts, internal salaries, gross margins	Restricted to authorized finance and leadership analytical groups
Internal Operational	Page view logs, web clickstreams, service latencies	Accessible to all internal engineering and data teams
Public	Public marketing website blog posts, published whitepapers	No access restrictions

CHAPTER 9: DATA & ANALYTICS LEXICON: 40 ESSENTIAL INDUSTRY TERMS

The essential vocabulary required to speak the language of analytics.

Mastering analytics terminology enables seamless collaboration with data engineers, executives, and product leaders. Learn these 40 terms thoroughly.

FIELD STANDARD: METRIC DEFINITION DISCIPLINE

Never use the word 'Churn' without clarifying whether it refers to Customer Logo Churn (% of customers lost) or Net Revenue Churn (% of MRR lost).

Term	Industry Definition & Practical Engineering Context
Grain	The level of detail represented by a single row in a table (e.g. one row per transaction vs one row per customer per day).
Dimensional Modeling	A design technique for data warehouses using Fact and Dimension tables to optimize for analytical queries.
Star Schema	A database structure consisting of one central fact table referencing multiple surrounding dimension tables.
dbt (Data Build Tool)	A transformation tool that allows analysts to write data pipeline queries as modular SQL models with version control.
Window Function	A SQL function that performs calculations across a set of table rows that are related to the current row without collapsing them.
Cohort Analysis	A study that groups users by a shared characteristic (e.g. signup month) to track behavioral trends and retention over time.
Sample Ratio Mismatch (SRM)	A statistical check verifying that an A/B test split (e.g. 50/50) matches the actual traffic received without assignment bias.
Data Lineage	The lifecycle path that tracks data from its raw ingestion origin, through transformations, to its final dashboard destination.
ELT vs ETL	Extract-Load-Transform (modern pattern loading raw data directly into the warehouse before transforming) vs Extract-Transform-Load.
Slowly Changing Dimension (SCD)	A methodology used to track and preserve historical changes to dimension records over time.

CHAPTER 10: THE FIRST 90 DAYS SURVIVAL GUIDE FOR JUNIOR DATA ANALYSTS

Your roadmap for mastering the company data model, building trust, and driving decisions.

Your first 90 days as a data analyst are about building deep domain intuition, understanding upstream quirks in raw data, and earning trust by verifying every query total against official finance reports before publishing dashboards.

FIELD STANDARD: THE SANITY CHECK RULE

Before sharing any dashboard or query result with stakeholders, always sanity check the numbers: Does the total match the company billing report? Do row counts look physically plausible?

Timeline Milestone	Key Objective & Deliverable	Success Metric
Week 1: Data Access & Warehouse Setup	Obtain warehouse permissions; configure dbt profile and BI tool access	Executed first SELECT query against staging data warehouse
Week 2: Shadowing & Entity Learning	Map out the core entities: Users, Orders, Products, and Payments	Created an entity relationship sketch; understood primary keys and foreign joins
Day 30: First Business Analysis	Answer a prioritized ad-hoc stakeholder question with SQL and documentation	Delivered clear, verified findings with zero arithmetic or join errors
Day 60: dbt Model Contribution	Author a new dbt mart model with schema tests and documentation	PR reviewed and merged into the production analytics repository
Day 90: Executive Dashboard Launch	Build and publish an automated executive KPI dashboard for product or marketing	Dashboard adopted by leadership for weekly business review meetings